

8-bit CPU

[16-bit instruction]

OP codes

0000 = ADD

0001 = SUB

0010 = LOAD

0011 = STORE

0100 = JUMP

Bit format ① ADD/SUB

Ex:

[15:12] → opcode

ADD R₁, R₂

[11:10] → rd

SUB R₁, R₂

[9:8] → rs

[7:0] unused

② LOAD

Ex:

[15:12] → opcode

LOAD R₁, [0x20]

[11:12] → rd

[9:8] → unused

[7:0] → memory address

③ STORE

Ex:

[15:12] → opcode

STORE R2, 0x10

[11:10] → rs

[9:8] → unused

[7:0] → memory address

④ JUMP

Ex:

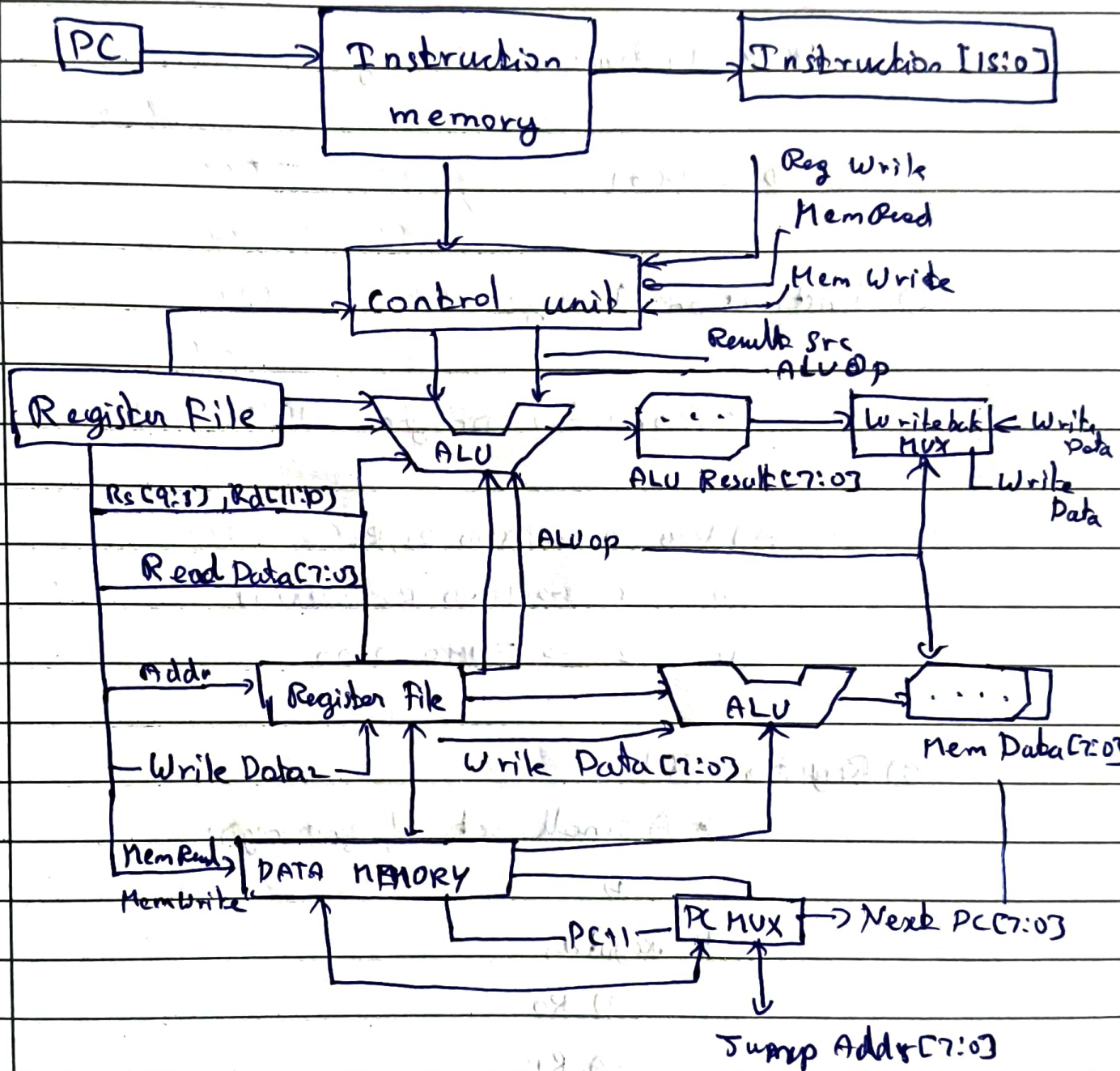
[15:12] → opcode

JUMP 0x10

[11:8] → unused

[7:0] → jump address

DATA PATH



Block role:

① Program counter (PC)

PC $\rightarrow$ Instruction Memory

PC $\leftarrow$ PC + 1

② Instruction Memory:

* stores the program [PC reads]

Address 0 $\rightarrow$ ADD R₁, R₂

" 1 $\rightarrow$ LOAD R₃, COX207

" 2 $\rightarrow$ JUMP 0X00

③ Register File

* A small set of fast registers

④

4 Registers.

① R₀

② R₁

③ R₂

④ R₃

④ ALU

Perform ADD

② SUB

⑤ DATA MEMORY

* LOAD and STORE

1) LOAD → memory output goes to register

2) STORE → register value goes to memory

⑥ Control Unit

* Reads opcode and generates control signal

⑦

5) Control signals:

- * RegWrite = write to register file
- * MemRead = read data memory
- * MemWrite = write data memory
- * ALUOp = ADD or SUB
- * ResultSrc = ~~ALU~~ ALU result or memory data
- * PCSrc = PC+1 or jump address

6) Control Signals Table

Instruction	RegWrite	MemRead	MemWrite	ResultSrc	PCSrc	ALUSrc
ADD	1	0	0	0	0	ADD
SUB	1	0	0	0	0	SUB
LOAD	1	1	0	0	0	-
STORE	0	0	1	X	0	-
JUMP	0	0	0	X	1	X

X → do, 1b core

⑦ Forch instruction flow:

① ADD R_1, R_2

PC $\rightarrow$ instruction memory

instruction $\rightarrow$ control unit

R_1 and $R_2 \rightarrow$ register file outputs

R_1 and $R_2 \rightarrow$ ALU

ALU results $\rightarrow$ write back MUX

write back MUX $\rightarrow R_1$

PC $\rightarrow$ PC + 1

② SUB R_1, R_1

Same as ADD, but ALU subtracts

③ LOAD R1, [0x20]

PC $\rightarrow$ instruction memory

instruction $\rightarrow$ control unit

0x20 $\rightarrow$ data memory address

data memory output $\rightarrow$ write back MUX

Write

Writeback MUX $\rightarrow$ R1

PC $\rightarrow$ PC+1

④ STORE R2, [0x20]

PC $\rightarrow$ instruction memory

instruction $\rightarrow$ control unit

R2 $\rightarrow$ data memory input

0x20 $\rightarrow$ data memory address

PC = PC+1

⑤ JUMP op

PC $\rightarrow$ instruction memory

instruction memory $\rightarrow$ control unit

jump address $\rightarrow$ PC Mux

PC Mux $\rightarrow$ PC